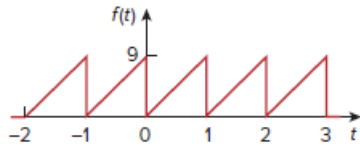


Problem

Determine the Fourier series of the sawtooth waveform in Fig. 17.9.

Figure 17.9:



Step-by-step solution

Step 1 of 5

Refer to waveform in Figure 17.9 in the textbook.

The function for sawtooth waveform is,

$$f(t) = 9t \quad 0 < t < 1 \text{ s}$$

Here,

Time period is $T = 1 \text{ s}$

Fundamental frequency is,

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{1} \\ &= 2\pi \end{aligned}$$

The trigonometric Fourier series of a function $f(t)$ is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega_0 t + b_n \sin n\omega_0 t]$$

Here,

$$a_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$a_n = \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt$$

Step 2 of 5

Calculate the coefficient a_0 .

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_0^T f(t) dt \\
 &= \int_0^1 9t dt \\
 &= 9 \left[\frac{t^2}{2} \right]_0^1 \\
 &= 9 \left[\frac{1^2 - 0}{2} \right] \\
 &= 4.5
 \end{aligned}$$

Step 3 of 5

Calculate the coefficient a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{1} \int_0^1 9t \cos(2\pi n t) dt
 \end{aligned}$$

It is known that,

$$\int t \cos(nt) dt = \frac{1}{n^2} \cos(nt) + \frac{1}{n} t \sin(nt)$$

Therefore,

$$\begin{aligned}
 a_n &= 18 \left[\frac{1}{(2\pi n)^2} \cos(2\pi n t) + \frac{t}{2\pi n} \sin(2\pi n t) \right]_0^1 \\
 &= 18 \left[\frac{1}{(2\pi n)^2} [\cos(2\pi n) - \cos(0)] + \frac{(1)}{2\pi n} \sin(2\pi n) - 0 \right] \\
 &= 18 \left[\frac{1}{(2\pi n)^2} [1 - 1] + 0 - 0 \right] \\
 &= 0
 \end{aligned}$$

Step 4 of 5

Calculate the coefficient b_n .

$$\begin{aligned} b_n &= \frac{2}{T} \int_0^T f(t) \sin(n\omega_0 t) dt \\ &= \frac{2}{1} \int_0^T 9t \sin(2\pi n t) dt \end{aligned}$$

It is known that,

$$\int t \sin(nt) dt = \frac{1}{n^2} \sin(nt) - \frac{1}{n} t \cos(nt)$$

Therefore,

$$\begin{aligned} a_n &= 18 \left[\frac{1}{(2\pi n)^2} \sin(2\pi n t) - \frac{t}{2\pi n} \cos(2\pi n t) \right]_0^1 \\ &= 18 \left[\frac{1}{(2\pi n)^2} [\sin(2\pi n) - \sin(0)] - \frac{(1)}{2\pi n} \cos(2\pi n) - 0 \right] \\ &= 18 \left[\frac{1}{(2\pi n)^2} [0 - 0] - \frac{1}{2\pi n} - 0 \right] \\ &= -\frac{9}{\pi n} \end{aligned}$$

Step 5 of 5

Calculate the Fourier series of the sawtooth waveform.

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty} [a_n \cos n\omega_0 t + b_n \sin n\omega_0 t] \\ &= 4.5 + \sum_{n=1}^{\infty} \left[(0) \cos 2\pi n t + \left(-\frac{9}{\pi n} \right) \sin 2\pi n t \right] \\ &= 4.5 + \sum_{n=1}^{\infty} \left[0 - \frac{9}{\pi n} \sin 2\pi n t \right] \\ &= 4.5 - \frac{9}{\pi} \sum_{n=1}^{\infty} \left[\frac{1}{n} \sin 2\pi n t \right] \end{aligned}$$

Therefore, the Fourier series expansion of the sawtooth waveform, $f(t)$ is

$$\boxed{4.5 - \frac{9}{\pi} \sum_{n=1}^{\infty} \left[\frac{1}{n} \sin 2\pi n t \right]}.$$